

INTERNATIONAL GCSE PHYSICS

9203/1

Paper 1

Mark scheme

June 2024

Version: 1.0 Final



2 4 6 Y 9 2 0 3 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 2 with a small amount of level 3 material it would be placed in level 2 but be awarded a mark near the top of the level because of the level 3 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives, level of demand and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

- 2.1** In a list of acceptable answers where more than one mark is available ‘any **two** from’ is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.
- 2.2** A bold **and** is used to indicate that both parts of the answer are required to award the mark.
- 2.3** Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.
- 2.4** Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that ‘right + wrong = wrong’.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols/formulae

If a student writes a chemical symbol/formula instead of a required chemical name, full credit can be given if the symbol/formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. At any point in a calculation students may omit steps from their working. If a subsequent step is given correctly, the relevant marks may be awarded.

Full marks should be awarded for a correct numerical answer, without any working shown. Full marks are not awarded for a correct final answer from incorrect working.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(.....) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, 'allow' is used to indicate creditworthy alternative answers.

3.9 Ignore

Ignore is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do not accept

Do **not** accept means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	stationary		1	AO1 3.1.2a 1–3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	constant speed		1	AO1 3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	25 seconds		1	AO2 3.1.2a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	distance = 15 (m)		1	AO2 3.1.2b
	speed = $\frac{15}{10}$		1	
	speed = 1.5 (m/s)		1	
		allow $\pm \frac{1}{2}$ a small square for any correct pair of values from the graph between A and B		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	the gradient would be greater	allow the line will be steeper	1	AO2 3.1.2b

Total Question 1		7
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Question 2

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	the oscillations are parallel to the direction of energy transfer	allow the oscillations are parallel to the direction of wave travel	1	AO1 3.3.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	0.004 s		1	AO2 3.3.3g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	$f = \frac{1}{0.004}$	ecf from Question 02.2	1	AO2 3.3.1g
	$f = 250$ (Hz)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	wave drawn with the same frequency		1	AO2 3.3.1g
	wave drawn with a smaller amplitude		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	$1500 = 20 \times \lambda$		1	AO2 3.3.1h
	$\lambda = \frac{1500}{20}$		1	
	$\lambda = 75$ (m)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6	1500 (m)		1	AO2 3.3.1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	hearing and smell		1	AO3 3.3.1

Total Question 2		11
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Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	the length of the wire		1	AO4 3.5.1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	the resistance of the wire		1	AO4 3.5.1

Question	Answers	Mark	AO/ Spec. Ref.
03.3	Current Ammeter Joulemeter Metre rule Potential difference Power pack Voltmeter do not accept more than one line from a box on the left	1 1	AO4 3.5.1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	5 points plotted correctly	allow 1 mark for 3 or 4 points plotted correctly	2	AO2 3.5.1
	line of best fit		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.5	directly proportional	allow 1 mark for as the length of the wire increases the resistance increases or allow 1 mark for positive correlation	2	AO3 3.5.1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.6	$12 = 0.20 \times R$		1	AO2 3.5.1h
	$R = \frac{12}{0.20}$		1	
	$R = 60 (\Omega)$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.7	electrons collide more often with ions		1	AO1 3.5.1 i m t
	which transfer energy and increase (amplitude of ion) vibration		1	
	increasing temperature therefore increasing resistance		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.8	the movable contact lengthens the wire so resistance increases		1	AO3 3.5.1
	therefore the current decreases (making the lamp less bright)	allow smaller potential difference across the lamp if no other mark awarded allow 1 mark for slide the moveable contact to the right to make the wire in the circuit longer	1	

Total Question 3		17
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Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	atoms with the same number of protons		1	AO1 3.7.1f
	but a different number of neutrons		1	
		if no other mark awarded allow 1 mark for same atomic number and different mass number		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	two protons and two neutrons		1	AO1 3.7.2e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	most ionising type of radiation		1	AO1 3.7.2g
	stopped by paper		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	there is friction between the paper and rollers		1	1 × AO1 1 × AO2 1 × AO3 3.5.1b
	electrons are transferred from the rollers to the paper		1	
	as paper is an insulator (there is a build up of charge)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	an atom that has lost or gained (outer) electron(s)	allow an atom that has lost electron(s) allow an atom that has gained electron(s)	1	AO1 3.7.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	positively charged ions attract the electrons from the paper		1 1	AO1 3.5.1b

Total Question 4		11
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Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	particles touching but randomly arranged		1	AO1 3.4.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	$\Delta\theta = 85\ (^{\circ}\text{C})$		1	AO2 3.4.1b
	$E = 0.60 \times 4200 \times 85$	allow correct substitution using an incorrect $\Delta\theta$	1	
	$E = 214\ 200\ (\text{J})$	allow 210 000 (J) allow correct calculation using an incorrect $\Delta\theta$	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	particles are much further apart		1	AO1 3.4.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	$E = 0.175 \times 2\ 300\ 000$		1	AO2 3.4.1d
	$E = 402\ 500$	allow a correct calculation using an incorrectly / not converted value of m	1	
	$E = 403\ 000\ (\text{J})$		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	particles with the greatest speed	allow (kinetic) energy for speed throughout	1	AO1 3.4.2 b
	leave the surface of the water		1	
	therefore the slower particles remain (in the water)		1	
	the lower the mean speed of the particles the lower the temperature (of the water)		1	

Total Question 5		12
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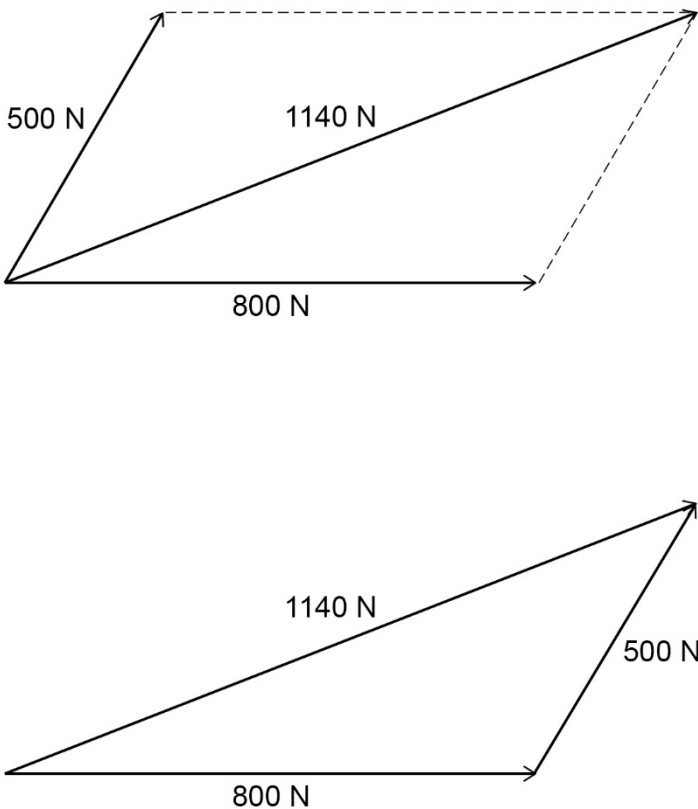
Question 6

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.1	a single force that has the same effect (on the motion) as all the individual forces	allow the combined effect of all the forces acting on an object	1	AO1 3.1.3g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	zero		1	AO1 3.1.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.3	$2\,700 = 900 \times a$	allow 3.0	1	3 × AO2 1 × AO1 3.1.3d
	$a = \frac{2\,700}{900}$		1	
	$a = 3$		1	
	m/s ²		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.4	the forces are equal in magnitude		1	AO1 3.1.3a
	and opposite in direction		1	
	due to Newton's 3 rd law		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.5	forces drawn to scale with the correct angle between them.		1	AO2 3.1.3b
	resultant force drawn		1	
	1140(N)	allow between 1090 and 1190	1	
				

Total Question 6		12
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Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	wider base		1	AO1 3.1.7c
	lower centre of mass		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	the total momentum before a collision is equal to the total momentum after the collision		1	2 × AO1 2 × AO3 3.1.4b
	so momentum of the footballer decreases		1	
	and momentum of the ball increases		1	
	and because momentum = mass × velocity, the velocity increases	allow a resultant force acts (so the ball accelerates) MP4 dependent on MP3	1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	$0.0090 = 0.0015 \times a$	allow incorrect calculated value of a for subsequent marks	1	AO2 3.1.3d 3.1.3h 3.1.4a 3.1.4c
	$a = 6 \text{ (m/s}^2\text{)}$		1	
	$6 = \frac{v}{0.12}$		1	
	$v = 6 \times 0.12$		1	
	$v = 0.72 \text{ (m/s)}$		1	
	OR	allow incorrect calculated value of Δp for subsequent marks		
	$\Delta p = 0.0090 \times 0.12 \text{ (1)}$			
	$\Delta p = 0.00108 \text{ (kg m/s) (1)}$			
	$0.00108 = 0.0015 \times v \text{ (1)}$			
	$v = \frac{0.00108}{0.0015} \text{ (1)}$			
	$v = 0.72 \text{ (m/s) (1)}$			

Total Question 7		11
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Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	step-down transformer		1	AO1 3.6.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	the power of the primary coil is the same as the power of the secondary coil		1	AO1 3.6.2f
	as power = current $\times$ potential difference		1	
	a step-down transformer reduces the potential difference		1	
	so therefore the current (in the secondary coil) must be greater		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	there is an alternating current in the coil in the card reader		1	AO3 3.6.2c
	causing an alternating magnetic field (around the coil)	allow changing for alternating for MP2	1	
	which induces an (alternating) potential difference across the coil in the bank card (which causes a current)		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.4	there is no power source that needs to be recharged or replaced		1	AO3 3.6.3

Total Question 8		9
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